Session 7 Mini Assignment

Source: Prompt Engineering Assignment
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SEC : C

Objective

Design a 3-step prompt chaining workflow: Step 1 uses few-shot prompting to extract entities, Step 2 uses Chain of Thought to determine urgency, and Step 3 outputs valid JSON with entities, urgency, and summary.

Support Ticket (Input)

Customer Raj reports that payment failed twice, money was deducted, and he needs the account restored before tomorrow.

Step 1 – Few-shot Prompt

Extract the key entities from a support ticket.

Example 1:
Ticket: Internet is down for Priya in Chennai.
Entities: [Priya, Internet, Chennai]

Example 2:
Ticket: Amit cannot log into his bank account.
Entities: [Amit, Bank Account, Login]

Now extract entities from:
Customer Raj reports that payment failed twice, money was deducted, and he needs the account restored before tomorrow.

Step 1 Output

- Raj
- Payment Failure
- Money Deducted
- Account Restoration
- Tomorrow

Step 2 – CoT Prompt

Use the extracted entities to reason step by step.

Rules:
- High: financial loss or urgent deadline
- Medium: service issue without immediate risk
- Low: general inquiry

Entities:
Raj, Payment Failure, Money Deducted, Account Restoration, Tomorrow

Return the urgency level and a one-line reason.

Step 2 Output

Reasoning:
1. Payment failed and money was deducted.
2. Customer has a deadline before tomorrow.
3. Financial impact + urgency = High.

Urgency: High

Step 3 – JSON Prompt

Convert the result into valid JSON only.

Schema:

```
{
  "entities": [...],
  "urgency": "high",
  "summary": "..."
}
```

Step 3 Output (Final JSON)

```
{
  "entities": [
    "Raj",
    "Payment Failure",
    "Money Deducted",
    "Account Restoration",
    "Tomorrow"
  ],
  "urgency": "high",
  "summary": "Raj reported a payment failure with money deducted and requires urgent account restoration before tomorrow."
}
```

End-to-End Workflow

Stage	Result
Input	Customer Raj reports that payment failed twice, money was deducted, and he needs the account restored before tomorrow.
Step 1	Entities extracted
Step 2	Urgency = High
Step 3	Valid JSON generated